Data Structures & Algorithms (DSA) Question Bank

Arrays

1. Find the largest and smallest element in an array.
2. Reverse an array in-place.
3. Find the second largest element.
4. Move all zeros to the end.
5. Find duplicates in an array.
6. Kadane's Algorithm for maximum subarray sum.
7. Merge two sorted arrays.
8. Rotate an array by k positions.
9. Find missing number in an array.
10. Two Sum problem.

Linked Lists

1. Reverse a linked list.
2. Detect a cycle in a linked list.
3. Find the middle node.
4. Merge two sorted linked lists.
5. Remove nth node from the end.
6. Check if linked list is palindrome.
7. Intersection of two linked lists.
8. Flatten a linked list.

Stacks and Queues

1. Implement stack using queues.
2. Implement queue using stacks.
3. Valid parentheses problem.
4. Next Greater Element.
5. Design Min Stack.
6. LRU Cache implementation.

Trees

1. Inorder, Preorder, and Postorder traversals.
2. Level order traversal.
3. Height of a binary tree.
4. Check if tree is balanced.
5. Lowest Common Ancestor.

6. Diameter of a binary tree.
7. Serialize and deserialize a binary tree.

Graphs

1. BFS traversal.
2. DFS traversal.
3. Detect cycle in graph.
4. Topological sorting.
5. Dijkstra's Algorithm.
6. Minimum Spanning Tree.
7. Shortest path in unweighted graph.

Dynamic Programming

1. Fibonacci using DP.
2. 0/1 Knapsack.
3. Longest Common Subsequence.
4. Longest Increasing Subsequence.
5. Coin Change problem.
6. Matrix Chain Multiplication.
7. Edit Distance.

Sorting & Searching

1. Implement Quick Sort.
2. Implement Merge Sort.
3. Binary Search.
4. Search in Rotated Sorted Array.
5. Count Inversions.
6. Kth Smallest Element.